Photovoltaic models

Sergi Costa

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Typical models

Listed below are the typical photovoltaic cell models found in the literature. More complex models may provide better accuracy, however require more data which is not always available on the module's datasheet. See an example of a datasheet in the appendix.

- Ideal model:
  The net cell current I is composed of the photocurrent (the current generated from the incoming irradiance) I_{Ph} and the current through the diode I_d. The photocurrent is constant for a certain irradiance, however the diode absorbs more current when its voltage increases, thus:

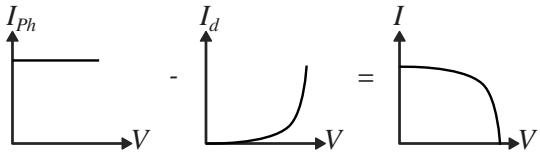


Figure 0.1: Characteristic current-voltage curve of the ideal cell model

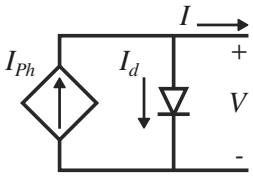


Figure 0.2: One diode ideal cell model

- 1 diode model with series resistance R_s:
  Assuming R_p is high enough ($R_p > 1$ kΩ)

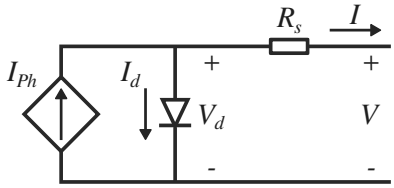


Figure 0.3: One diode R_s PV cell model

- 1 diode model with series R_s and shunt R_p resistances:

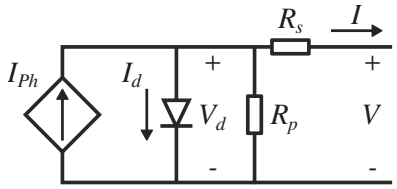


Figure 0.4: One diode R_s and R_p PV cell model

- 2 diode model:

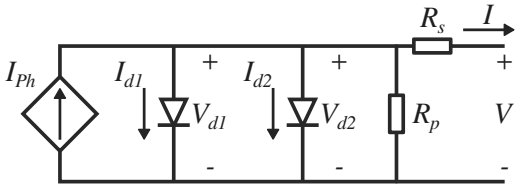


Figure 0.5: Two diode R_s and R_p PV cell model

Normalized conditions

As developed in [1]:

- Standard test conditions (STC)
  - irradiance $G_{STC} = 1000$ W/m^2
  - cell temperature $T_{STC} = 25$ ^{o}C
- Nominal operating cell temperature (NOCT)
  - irradiance $G_{NOCT} = 800$ W/m^2
  - ambient temperature $T_{amb} = 20$ ^{o}C
  - cell temperature $T_{NOCT} \approx 45$ - 50 ^{o}C
- Low irradiance condition (LIC)
  - irradiance $G_{STC} = 200$ W/m^2
  - cell temperature $T_{STC} = 25$ ^{o}C
- High temperature condition (HTC)
  - irradiance $G_{STC} = 1000$ W/m^2
  - cell temperature $T_{STC} = 75$ ^{o}C
- Low temperature condition (LTC)
  - irradiance $G_{STC} = 500$ W/m^2
  - cell temperature $T_{STC} = 15$ ^{o}C

Computing the temperature of the module T

The temperature of the cell T_c (or module T) usually differs from the ambient temperature T_{amb}. See implementations and other examples in the *pvlib* python library[1]. The following proposed models compute the actual temperature of the module:

- Ross cell temperature model [2]:

$$T_c = T_{amb} + k\,G \tag{0.1}$$

  T_{amb}: ambient temperature (K).
  G: irradiance (W/m^2).
  k: Ross coefficient (m^{2}K/W). = 0.02 (well cooled) to 0.0563 (sloped roof, non-ventilated) [3].

[1] pvlib-python.readthedocs.io

- Linear temperature model [4, Ch5.4]:

$$T = T_{amb} + c\,\frac{G}{G_{STC}} \tag{0.2}$$

  T_{amb}: ambient temperature (K).
  c: constant for the linear model. 20 ^oC for free standing modules, 30 ^oC for roof mounted with ventilation space, 45 ^oC for roof mounted without ventilation space.
  G_{STC}: irradiance at STC (W/m^2).
  G: irradiance (W/m^2).

- Sandia module temperature model [5]:

$$T = G\,(\exp(a + b\,w)) + T_{amb} \tag{0.3}$$

  G: irradiance (W/m^2).
  T_{amb}: ambient temperature (^{o}C).
  w: wind speed (m/s).
  a, b: module parameters (see Table 0.1)

- Sandia cell temperature model [5]:

$$T_c = T + \frac{G}{G_{STC}}\,\Delta T \tag{0.4}$$

  T: module temperature (^{o}C).
  G: irradiance (W/m^2).
  G_{STC}: irradiance at STC (W/m^2).
  ΔT: Temperature difference between cell and module at $G = 1000$ W/m^2 (see Table 0.1).

Type	Mount	a	b	ΔT
Glass/cell/glass	Open rack	-3.47	-0.0594	3
Glass/cell/glass	Close roof	-2.98	-0.0471	1
Glass/cell/poly.	Open rack	-3.56	-0.0750	3
Glass/cell/poly.	Insulated back	-2.81	-0.0455	0
Poly./thin-film/steel	Open rack	-3.58	-0.113	3
22X linear conc.	Tracker	-3.23	-0.130	13

Table 0.1: Sandia model empirical parameters

- Faiman temperature model [6, 7]:

$$T = T_{amb} + \frac{G}{U_0 + U_1\,w} \tag{0.5}$$

T_{amb}: ambient temperature (^{o}C).
G: irradiance (W/m^2).
w: wind speed (m/s).
U_0, U_1: constant and convective heat transfer coefficients (see Table 0.2) in (W/(m^2 K)) and (W/(m^3 K)).

Ref.	U_0	U_1
Negev desert [6]	23.5 - 26.4	6.25 - 7.68
Cologne, DE [7]	35.7	8.22
Ancona, IT [7]	41.9	3.95
Tempe, US [7]	32.1	6.08
Thuwai, SA [7]	39.7	3.06
Chennai, IN [7]	30.1	4.75

Table 0.2: Constant and convective heat transfer coefficients

- NOCT (Messenger et al.) cell temperature model [8, Ch.3]:

$$T_c = T_{amb} + \frac{T_{NOCT} - 20}{800} G \qquad (0.6)$$

- NOCT cell temperature model: TODO https://pvpmc.sandia.gov/modeling-guide/2-dc-module-iv/cell-temperature/noct-cell-temperature/
- Fuentes cell temperature model: TODO https://pvpmc.sandia.gov/modeling-guide/2-dc-module-iv/cell-temperature/fuentes-cell-temperature/
- Dynamic temperature model with a first order transfer function [9, 10]:

$$T = T_{amb} + T_0 + \frac{k}{1 + \tau\, s} G \qquad (0.7)$$

T_{amb}: ambient temperature (K).
T_0: correction temperature = -3 $^o C$.
k: irradiance gain = 0.03 K/(W m^2).
τ: module time constant = 5 min.
G: irradiance (W/m^2).

- Dynamic temperature model from thermal balance [4, Ch5.4]: TODO

1 Imperial College ELEC 40009 model

This model is presented in the ELEC 40009 course.

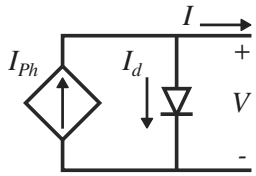


Figure 1.1: One diode ideal PV cell model

The photovoltaic cell is equivalent to a current source in parallel to a diode (Figure 1.1).

For a single module or cell:

- Current balance (Kirchhoff current law):

$$I = I_{Ph} - I_d \tag{1.1}$$

- Current generated by incident light I_{Ph}:

$$I_{Ph} = I_{sc,STC} \frac{G}{G_{STC}} \tag{1.2}$$

  $I_{sc,STC}$: short circuit current at STC (A).
  G_{STC}: irradiance at STC (W/m^2).
  G: irradiance (W/m^2).

- Shockley diode equation I_d:

$$I_d = I_0 \left(\exp\left(\frac{V}{\alpha_d V_t} \right) - 1 \right) \tag{1.3}$$

  I_0: reverse saturation current or leakage current of the diode ($1{\cdot}10^{-10}$ A @ 298 K (25 ^oC), doubles each +5 ^oC).
  α_d: diode ideality factor (1 - 2, usually $1 \leq \alpha_d \leq 1.5$).
  V_t: thermal voltage ($\approx$ 25 mV @ 298 K (25 ^oC).

- Thermal voltage V_t:

$$V_t = \frac{k\,T}{q} \tag{1.4}$$

  k: Boltzmann constant $1.3807{\cdot}10^{-23}$ J/K.
  q: elementary charge (charge of an electron) $1.6022{\cdot}10^{-19}$ C.
  T: temperature of the cell (K).

- Reverse saturation current I_0:

$$I_0 = I_{0,STC} \cdot 2^{(T - T_{STC})/5} \tag{1.5}$$

  $I_{0,STC}$: reverse saturation current at 25 ^oC = $1{\cdot}10^{-10}$ A.
  T_{STC}: = 25 ^oC.

Modules in series (N_s) and parallel (N_p):

$$V = V_{global}/N_s \tag{1.6}$$

$$I_{global} = I\,N_p \tag{1.7}$$

Imperial College ELEC 40009 model example

```
function I = fcn(T,G,V)

Gstc = 1e3;
Tstc = 25+273;

% Number of modules series / parallel
Ns = 1;
Np = 1;

% Number of cells per module
Ncells = 144/2;
Ncellp = 2;

% Cell parameters
a = 1.1;
k = 1.3807e-23; % J/K
q = 1.6022e-19; % C
Is0 = 1e-10; % A at 25deg
Iph_sc  = 13.86/Ncellp;

Vcell = Vdc/Ns/Ncells;
Iph = Iph_sc*(G/Gstc);
Is = Is0*2.^((T-Tstc)./5);
Id = Is*(exp(Vcell*q/(a*k*(T+273)))-1);
Icell = Iph - Id;
I = Icell*Ncellp*Np

I = max(I,0);
```

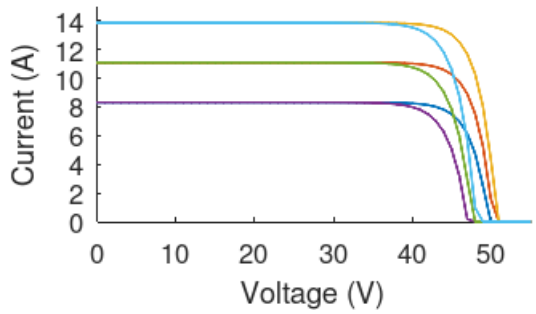


Figure 1.2: Current-voltage plot of the example module

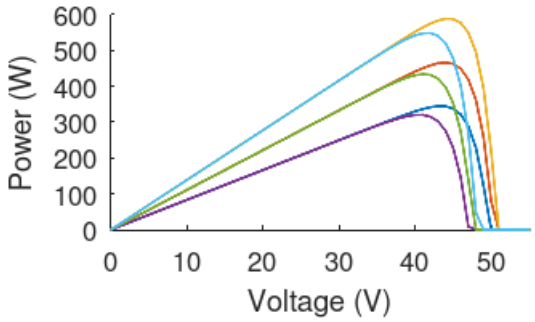


Figure 1.3: Power-voltage plot of the example module

2 EN-50530 model for MPPT performance test

This model is published in the Annex C of the EN-50530 European standard for MPPT performance tests[2] [10]. The model makes use only of parameters provided by manufacturers datasheets and technology depending factors which are provided by the standard itself.

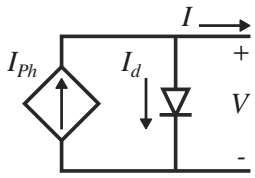


Figure 2.1: One diode ideal PV module model

The photovoltaic module is modelled as a current source in parallel to a diode (Figure 2.1).

For a single module:

- Characteristic curve:

$$I = I_{sc} - I_0 \left(\exp\left(\frac{V}{V_{oc}\, C_{AQ}} \right) - 1 \right) \tag{2.1}$$

- Short circuit current corrected I_{sc}:

$$I_{sc} = I_{sc,STC} \frac{G}{G_{STC}} \left(1 + K_I \left(T - T_{STC}\right)\right) \tag{2.2}$$

  $I_{sc,STC}$: short circuit current at STC (A).
  G_{STC}: irradiance at STC (W/m^2).
  G: irradiance (W/m^2).
  K_I: temperature coefficient of the current (%/^oC). = 0.04 %/^oC for cSi-technology and 0.02 %/^oC for thin film technology.
  T: temperature of the module (K).
  T_{STC}: temperature at STC (K).

- Open circuit voltage corrected V_{oc}:

$$V_{oc} = V_{oc,STC}\ \left(1 + K_V \left(T - T_{STC}\right)\right) \cdot \cdot \left(\ln\left(\frac{G}{C_G} + 1 \right) C_V - C_R\, G \right) \tag{2.3}$$

  $V_{oc,STC}$: open circuit voltage at STC (V).
  K_V: temperature coefficient of the voltage (%/^oC). = -0.4 %/^oC for cSi-technology and -0.2 %/^oC for thin film technology.
  T: temperature of the module (K).
  T_{STC}: temperature at STC (K).
  G: irradiance (W/m^2).
  C_G, C_V, C_R: technology depending correction factors. = 2.514·10^{-3} W/m^2, 8.593·10^{-2} and 1.088·10^{-4} m^2/W respectively for cSi-technology and 1.252·10^{-3} W/m^2, 8.419·10^{-2} and 1.476·10^{-4} m^2/W respectively for thin film technology.

- Saturation current corrected I_0:

$$I_0 = I_{sc,STC} \frac{G}{G_{STC}} \left(1 - FF_I\right)^{1/(1 - FF_V)} \tag{2.4}$$

- MPP to open circuit voltage ratio FF_V (Fill factor):

$$FF_V = \frac{V_{mpp,STC}}{V_{oc,STC}} \tag{2.5}$$

  $V_{mpp,STC}$: maximum power point voltage at STC (V).
  $V_{oc,STC}$: open circuit voltage at STC (V).

- MPP to short circuit current ratio FF_I (Fill factor):

$$FF_I = \frac{I_{mpp,STC}}{I_{oc,STC}} \tag{2.6}$$

  $V_{mpp,STC}$: maximum power point voltage at STC (V).
  $V_{oc,STC}$: open circuit voltage at STC (V).

- Constant C_{AQ}:

$$C_{AQ} = \frac{FF_V - 1}{\ln(1 - FF_I)} \tag{2.7}$$

[2] AENOR, "UNE-EN 50530:2011 Rendimiento global de los inversores fotovoltaicos," June 2011.

EN-50530 model for MPPT test example

```
function I = fcn(T,G,V)

Gstc = 1e3;
Tstc = 25+273;

% Number of modules series / parallel
Ns = 1;
Np = 1;

% Module parameters @ STC
Vmpp = 41.64;
Impp = 12.97;
Vocstc = 49.60;
Iscstc = 13.86;

% cSi technology
KI = 0.045/100;
KV = -0.275/100;
Cg = 2.514e-3;
Cv = 0.08593;
Cr = 1.088e-4;

FFv = Vmpp/Vocstc;
FFi = Impp/Iscstc;
Caq = (FFv-1)/log(1-FFi);

Isc = Iscstc*(G/Gstc)*(1+KI*(T-Tstc));
I0 = Iscstc*(G/Gstc)*(1-FFi)^(1/(1-FFv))
    ;
Voc = Vocstc*(1+KV*(T-Tstc))*(log((G/Cg)
    +1)*Cv-Cr*G);

Vmod = V/Ns;
Imod = Isc-I0*(exp(Vmod/(Voc*Caq))-1);
I = Imod*Np;

I = max(I,0);
```

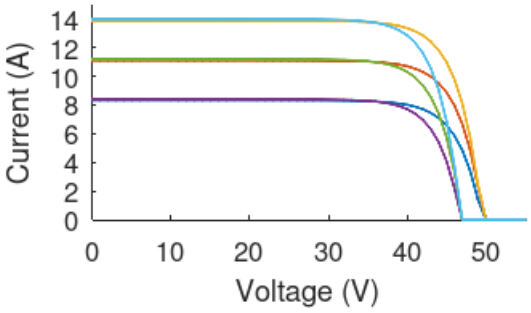


Figure 2.2: Current-voltage plot of the example module

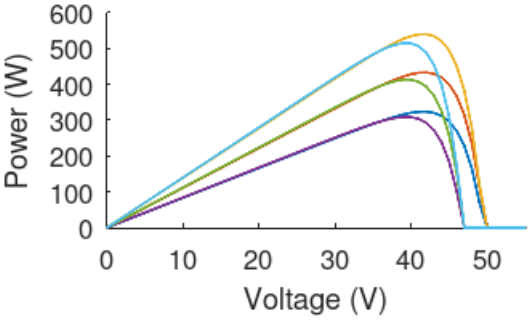


Figure 2.3: Power-voltage plot of the example module

3 (A.Bellini et al. model)

TODO: solve C1 and C2

This model is proposed by Bellini et al.[3], which is published in [11]. The proposed model makes use only of parameters provided by manufacturers' datasheets.

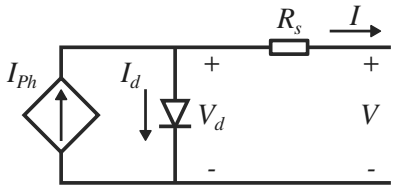


Figure 3.1: One diode R_s PV cell model

The photovoltaic module is modelled as a current source in parallel to a diode with a series resistor R_s (Figure 3.1).

For a singe cell:

- Characteristic I-V curve:

$$I = I_{sc}\left[1 - C_1\left(\exp\left(\frac{V}{C_2\,V_{oc}}\right) - 1\right)\right] \tag{3.1}$$

I_{sc}: short circuit current (A).
V_{oc}: open circuit voltage (V).

- Coefficient C_1:

$$C_1 = \left(1 - \frac{I_{mpp}}{I_{sc}}\right)\exp\left(\frac{-V_{mpp}}{C_2\,V_{oc}}\right) \tag{3.2}$$

I_{sc}: short circuit current (A).
V_{oc}: open circuit voltage (V).
I_{mpp}: maximum power point current (A).
V_{mpp}: maximum power point voltage (V).

- Coefficient C_2:

$$C_2 = \frac{V_{mpp}/V_{oc} - 1}{\ln\left(1 - I_{mpp}/I_{sc}\right)} \tag{3.3}$$

[3]A. Bellini, S. Bifaretti, V. Iacovone, and C. Cornaro, "Simplified model of a photovoltaic module," in 2009 Applied Electronics, pp. 47–51, 2009

I_{sc}: short circuit current (A).
V_{oc}: open circuit voltage (V).
I_{mpp}: maximum power point current (A).
V_{mpp}: maximum power point voltage (V).

- Short circuit current I_{sc}:

$$I_{sc} = I_{sc,STC}\frac{G}{G_{STC}}\left(1 + K_I\Delta T\right) \tag{3.4}$$

$I_{sc,STC}$: short circuit current at STC (A).
G_{STC}: irradiance at STC (W/m^2).
G: irradiance (W/m^2).
K_I: temperature coefficient of the current (%/^oC).
ΔT: actual to nominal cell temperature difference $\Delta T = T - T_{STC}$ (K).

- Open circuit voltage V_{oc}:

$$V_{oc} = V_{oc,STC}\left(1 + K_V\Delta T\right) - \Delta V \tag{3.5}$$

$V_{oc,STC}$: open circuit voltage at STC (V).
K_V: temperature coefficient of the voltage (%/^oC).
ΔT: actual to nominal cell temperature difference $\Delta T = T - T_{STC}$ (K).
ΔV: voltage correction term (V).

- Maximum power point current I_{mpp}:

$$I_{mpp} = I_{mpp,STC}\frac{G}{G_{STC}}\left(1 + K_I\Delta T\right) \tag{3.6}$$

$I_{mpp,STC}$: maximum power point current at STC (A).

- Maximum power point voltage V_{mpp}:

$$V_{mpp} = V_{mpp,STC}\left(1 + K_V\Delta T\right) - \Delta V \tag{3.7}$$

$V_{mpp,STC}$: maximum power point voltage at STC (V).

- Voltage correction term ΔV:

$$\Delta V = V_{oc,STC} - V_{OC,m} \tag{3.8}$$

$V_{oc,STC}$: open circuit voltage at STC (V).
$V_{oc,m}$: open circuit voltage irradiance translated (V).

- Translated open circuit voltage $V_{oc,m}$:

$$V_{oc,m} = C_2\, V_{oc.STC}\, \ln\left[1 + \frac{(1 - I_{sc,m}/I_{sc,STC}}{C_1}\right] \tag{3.9}$$

$I_{sc,m}$: short circuit current irradiance translated (A).

- Translated short circuit current $I_{sc,m}$:

$$I_{sc,m} = I_{sc,STC}\frac{G}{G_{STC}} \tag{3.10}$$

Modules in series (N_s) and parallel (N_p):

$$V = V_{global}/N_s \tag{3.11}$$

$$I_{global} = I\, N_p \tag{3.12}$$

$$R_{s,global} = R_s\, N_s \tag{3.13}$$

Equivalent series resistance R_s:

$$R_s = -C_2\left(\frac{V_{oc}}{I_{sc}}\right)\left(\frac{1}{1 + C_1}\right) \tag{3.14}$$

Bellini et al. model example

```
function I = fcn(T,G,V)

Gstc = 1e3;
Tstc = 25+273;
DT = T-Tstc;

% Number of modules series / parallel
Ns = 1;
Np = 1;

% Number of cells per module
Ncells = 144/2;
Ncellp = 2;


I = I*Ncellp*Np;
I = max(I,0);
```

4 W. De Soto et al. (Five-parameter) model

This model is proposed by De Soto et al.[4], which is published in [12]. The model requires five parameters at STC, being $V_{t,STC}$, $I_{0,STC}$, $I_{Ph,STC}$, $R_{s,STC}$ and $R_{p,STC}$.

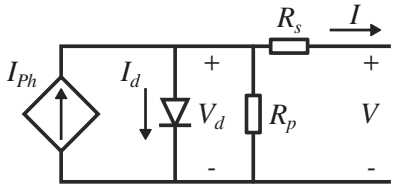


Figure 4.1: One diode R_s and R_p PV cell model

The photovoltaic module is modelled as a current source in parallel to a diode with a series R_s and a shunt resistor R_p (Figure 4.1).

For a singe cell:

- Current balance (Kirchhoff current law):

$$I = I_{Ph} - I_d - I_{R_p} \tag{4.1}$$

- Photocurrent I_{Ph}:

$$I_{Ph} = \frac{G}{G_{STC}} \frac{M}{M_{STC}} \left(I_{Ph,STC} + K_I \left(T - T_{STC}\right)\right) \tag{4.2}$$

G_{STC}: irradiance at STC (W/m^2).
G: irradiance (W/m^2).
M/M_{STC}: Air mass modifier ratio.
$I_{Ph,STC}$: photocurrent at STC (A).
K_I: temperature coefficient of the current (%/^oC).
T_{STC}: temperature of the cell at STC (K).
T: temperature of the cell (K).

- Air mass modifier M:

$$\frac{M}{M_{STC}} = \sum_0^4 a_i \left(AM\right)^i \tag{4.3}$$

a_i: material constants. See Table 4.1).

	Si thin	Single-cr.	Poly-cr.	3 junc. am.
a_0	0.938110	0.935823	0.918093	1.10044085
a_1	0.062191	0.054289	0.086257	-0.06142323
a_2	-0.015021	-0.008677	-0.024459	-0.00442732
a_3	0.001217	0.000527	0.002816	0.000631504
a_4	-0.000034	-0.000011	-0.000126	-1.9184E-05
α_d	1.357	1.026	1.025	3.09
$E_{g,STC}$	1.12	1.12	1.14	1.6

Table 4.1: Material parameters [12]

- Air mass AM [13]:

$$AM = \frac{1}{\cos(\theta_z) + 0.5057 \left(96.08 - \theta_z\right)^{-1.634}} \tag{4.4}$$

θ_z: zenith angle (^{o}).

- Shockley diode equation I_d:

$$I_d = I_0 \left(\exp\left(\frac{V_d}{\alpha_d V_t}\right) - 1\right) \tag{4.5}$$

I_0: reverse saturation current or leakage current of the diode (A).
α_d: diode ideality factor ($1 \leq \alpha_d \leq 1.5$). See Table 4.1.
V_t: thermal voltage. (!) if computing for a module should be multiplied by the number of cells in series $N_{cell,s}$.

- Reverse saturation current I_0:

$$I_0 = I_{0,STC} \left(\frac{T}{T_{STC}}\right)^3 \exp\left[\frac{1}{k}\left(\frac{E_{g,STC}}{T_{STC}} - \frac{E_g}{T}\right)\right] \tag{4.6}$$

$I_{0,STC}$: Reverse saturation at STC (A).
E_g: Material bandgap (eV).
$E_{g,STC}$: Material bandgap at STC (eV). See Table 4.1.
T: temperature of the cell (K).
T_{STC}: temperature of the cell at STC (K).
k: Boltzmann constant 8.617333262·10^{-5} eV/K.

- Bandgap E_g:

$$E_g = E_{g,STC} \left(1 - \frac{dEg}{dT}(T - T_{STC})\right) \tag{4.7}$$

[4]W. De Soto, S. Klein, and W. Beckman, "Improvement and validation of a model for photovoltaic array performance," Solar Energy, vol. 80, pp. 78–88, Jan. 2006, DOI: 10.1016/j.solener.2005.06.010

$E_{g,STC}$: Material bandgap at STC (eV). See Table 4.1.
dEg/dT: Bandgap temperature dependence (eV/T). = 0.0002677 for Si.
T: temperature of the cell (K).
T_{STC}: temperature of the cell at STC (K).

- Thermal voltage V_t:

$$V_t = \frac{k\,T}{q} \tag{4.8}$$

k: Boltzmann constant $1.3807{\cdot}10^{-23}$ J/K.
q: elementary charge (charge of an electron) $1.6022{\cdot}10^{-19}$ C.
T: temperature of the cell (K).

- Diode voltage V_d:

$$V_d = V + R_s\,I \tag{4.9}$$

- Parallel resistance current I_{R_p}:

$$I_{R_p} = \frac{V_d}{R_p} \tag{4.10}$$

- Effective parallel resistance R_p:

$$R_p = \frac{G_{STC}}{G} R_{p,STC} \tag{4.11}$$

Modules in series (N_s) and parallel (N_p):

$$V = V_{global}/N_s \tag{4.12}$$

$$I_{global} = I\,N_p \tag{4.13}$$

$$R_{s,global} = R_s\,N_s \tag{4.14}$$

$$R_{p,global} = R_p\,N_s \tag{4.15}$$

De Soto et al. model example

```
function I = fcn(T,G,V)

M = 1.5; % assume Mstc

Gstc = 1e3;
Tstc = 25+273;
DT = T-Tstc;
Mstc = 1.5;

% Number of modules series / parallel
Ns = 1;
Np = 1;

% Number of cells per module
Ncells = 144/2;
Ncellp = 2;

% Cell parameters
a = 1.026;
k = 1.3807e-23; % J/K
k_evK = 8.6617333262e-5; % eV/K
q = 1.6022e-19; % C
Rs = 0.2/Ncells; % Rs cell
Rpstc = 500/Ncells; % Rp cell
Rp = Rpstc*Gstc/G;
Egstc = 1.12;
dEgdT = 0.0002677;
I0stc = 1e-10; % assume

Vocstc = 49.60/Ncells;
Iscstc = 13.86/Ncellp;
KI = 0.045/100;
KV = -0.275/100;

Eg = Egstc*(1-dEgdT*DT);

Iphstc = Iscstc; % assume
Iph = G/Gstc*M/Mstc*Iphstc*(1+KI*DT);

I0 = I0stc*(T/Tstc)^3*exp((Egstc/(k_evK*
    Tstc)-Eg/(k_evK*T)));
Vt = k*T/q;

% Iterative solution
Vcell = V/Ncells/Ns;
e=1e6; % error
I=0; % init
while e >= 1e-6
```

```
    Iprev = I;
    Vd = Vcell+I*Rs;
    Id = I0*(exp(Vd/(a*Vt))-1);
    I = Iph-Id-Vd/Rp;
    e = abs(I-Iprev);
end
I = I*Ncellp*Np;
I = max(I,0);
```

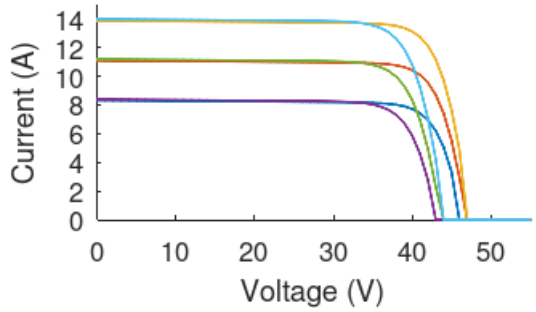


Figure 4.2: Current-voltage plot of the example module

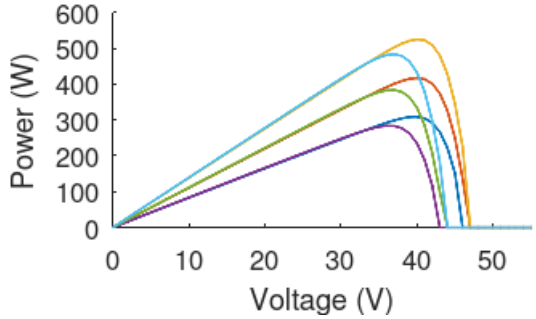


Figure 4.3: Power-voltage plot of the example module

5 (Alternative EN-50530 model)

TODO: esperar correu - no funciona

This model is published in the Annex C of the EN-50530 European standard[5] [10].

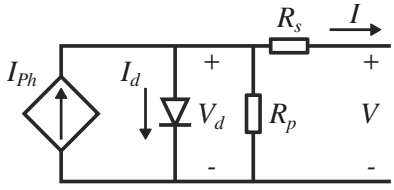


Figure 5.1: One diode with R_s and R_p PV module model

The photovoltaic module is modelled as a current source in parallel to a diode with a series R_s and a shunt resistor R_p (Figure 5.1) [10].

For a single module:

- Characteristic curve:

$$I = I_{Ph} - I_0 \left(\exp\left(\frac{V + I\,R_s}{\alpha_d\,V_t} \right) - 1 \right) - \frac{V + I\,R_s}{R_p} \tag{5.1}$$

  I_0: reverse saturation current or leakage current of the diode (A).
  α_d: diode ideality factor ($1 \leq \alpha_d \leq 1.5$). $= 1.113$ for crystalline technology.
  V_t: thermal voltage (V).
  R_s: Series resistance of the module (Ω). $= 47.731$ mΩ for crystalline technology.
  R_p: Parallel resistance of the module (Ω). $= 11.173\ \Omega$ for crystalline technology.

- Photocurrent I_{Ph}:

$$I_{Ph} = I_{Ph,STC}\,\frac{G}{G_{STC}} \tag{5.2}$$

  $I_{Ph,STC}$: photo current at G_{STC} (A). $= 1$ A for crystalline technology.
  G_{STC}: irradiance at STC (W/m^2).
  G: irradiance (W/m^2).

- Saturation current of the diode I_0:

$$I_0 = C_0\,T^3 \exp\left(\frac{-E_g}{k\,T_{amb}} \right) \tag{5.3}$$

  C_0: coefficient of I_0 (A/K^3). $= 101.668$ A/K^3 for crystalline technology.
  T: temperature of the module (K).
  T_{amb}: ambient temperature (K).
  E_g: Band gap (eV). $= 1.1$ eV for Si.
  k: Boltzmann constant $1.3807{\cdot}10^{-23}$ J/K.

- Thermal voltage V_t:

$$V_t = \frac{k\,T}{q} \tag{5.4}$$

  k: Boltzmann constant $1.3807{\cdot}10^{-23}$ J/K.
  q: elementary charge (charge of an electron) $1.6022{\cdot}10^{-19}$ C.
  T: temperature of the module (K).

Alternative EN-50530 model example

```
function I = fcn(T,G,V)

Gstc = 1e3;
Tstc = 25+273;

% Number of modules series / parallel
Ns = 1;
Np = 1;

% Number of cells per module
Ncells = 144/2;
Ncellp = 2;

% Cell parameters
Vocstc = 49.60/Ncells;
Iscstc = 13.86/Ncellp;

a = 1.113;
k = 1.3807e-23; % J/K
k_evK = 8.6617333262e-5; % eV/K
q = 1.6022e-19; % C
Rs = 47.731e-3; % Ohm
Rp = 11.173; % Ohm
Iphstc = 1*Iscstc;
C0 = 101.668;
Eg = 1.1; % eV
```

[5] AENOR, "UNE-EN 50530:2011 Rendimiento global de los inversores fotovoltaicos," June 2011.

```
Iph = Iphstc*G/Gstc;
I0 = C0*T^3*exp(Eg/(k_evK*Tamb));
Vt = k*T/q;

% iterative solution
Vcell = V/Ncells/Ns;
e = 1e6; % error
I = 0; % init
while e >= 1e-6
    Iprev = Ii;
    Vd = Vcell+I*Rs;
    I = Iph-I0*(exp(Vd/(a*Vt))-1)-Vd/Rp;
    e = abs(I - Iprev);
end
```

6 M.G.Villalva et al. model

This model is proposed by Villalva et al.[6], which is published in [14].

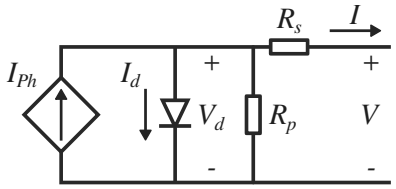


Figure 6.1: One diode R_s and R_p PV cell model

The photovoltaic module is modelled as a current source in parallel to a diode with a series R_s and a shunt resistor R_p (Figure 6.1).

For a singe cell:

- Current balance (Kirchhoff current law):

$$I = I_{Ph} - I_d - I_{R_p} \tag{6.1}$$

- Photocurrent I_{Ph}:

$$I_{Ph} = I_{Ph,STC}\,(1 + K_I\,\Delta T)\,\frac{G}{G_{STC}} \tag{6.2}$$

$I_{Ph,STC}$: photocurrent at STC (A). Either approximated as $I_{Ph,STC} \approx I_{sc,STC}$ or $I_{Ph,STC} = \dfrac{R_p + R_s}{R_p}\,I_{sc,STC}$.
K_I: temperature coefficient of the current ($\%/^oC$).
ΔT: actual to nominal cell temperature difference $\Delta T = T - T_{STC}$ (K).
G_{STC}: irradiance at STC (W/m^2).
G: irradiance (W/m^2).

- Shockley diode equation I_d:

$$I_d = I_0\left(\exp\left(\frac{V_d}{\alpha_d\,V_t}\right) - 1\right) \tag{6.3}$$

I_0: reverse saturation current or leakage current of the diode (A).
α_d: diode ideality factor ($1 \leq \alpha_d \leq 1.5$).
V_t: thermal voltage.

- Reverse saturation current I_0:

$$I_0 = \frac{I_{sc,STC}\,(1 + K_I\,\Delta T)}{\exp\left(V_{oc,STC}\,(1 + K_V\,\Delta T)/\left(\alpha_d\,V_t\right)\right) - 1} \tag{6.4}$$

$I_{sc,STC}$: short circuit current at STC (A).
K_I: temperature coefficient of the current ($\%/^oC$).
$V_{oc,STC}$: open circuit voltage at STC (V).
K_V: temperature coefficient of the voltage ($\%/^oC$).
ΔT: actual to nominal cell temperature difference $\Delta T = T - T_{STC}$ (K).

- Thermal voltage V_t:

$$V_t = \frac{k\,T}{q} \tag{6.5}$$

k: Boltzmann constant $1.3807{\cdot}10^{-23}$ J/K.
q: elementary charge (charge of an electron) $1.6022{\cdot}10^{-19}$ C.
T: temperature of the cell (K).

- Diode voltage V_d:

$$V_d = V + R_s\,I \tag{6.6}$$

- Parallel resistance current I_{R_p}:

$$I_{R_p} = \frac{V_d}{R_p} \tag{6.7}$$

Modules in series (N_s) and parallel (N_p):

$$V = V_{global}/N_s \tag{6.8}$$

$$I_{global} = I\,N_p \tag{6.9}$$

$$R_{s,global} = R_s\,N_s \tag{6.10}$$

$$R_{p,global} = R_p\,N_s \tag{6.11}$$

[6]M. Villalva, J. Gazoli, and E. Filho, "Comprehensive approach to modeling and simulation of photovoltaic arrays," IEEE Transactions on Power Electronics, vol. 24, pp. 1198–1208, May 2009, DOI: 10.1109/TPEL.2009.2013862

Villalva et al. model example

```
function I = fcn(T,G,V)

Gstc = 1e3;
Tstc = 25+273;
DT = T-Tstc;

% Number of modules series / parallel
Ns = 1;
Np = 1;

% Number of cells per module
Ncells = 144/2;
Ncellp = 2;

% Cell parameters
a = 1.113;
k = 1.3807e-23; % J/K
q = 1.6022e-19; % C
Rs = 0.2/Ncells; % Rs cell
Rp = 500/Ncells; % Rp cell

Vocstc = 49.60/Ncells;
Iscstc = 13.86/Ncellp;
KI = 0.045/100;
KV = -0.275/100;

Iphstc = (Rp+Rs)/Rp*Iscstc;
Iph = Iphstc*(1+KI*DT)*G/Gstc;
Vt = k*T/q;
I0 = Iscstc*(1+KI*DT)/(exp(Vocstc*(1+KV*
    DT)/(a*Vt))-1);

% Iterative solution
Vcell = V/Ncells/Ns;
e=1e6; % error
I=0; % init
while e >= 1e-6
    Iprev = I;
    Vd = Vcell+I*Rs;
    Id = I0*(exp(Vd/(a*Vt))-1);
    I = Iph-Id-Vd/Rp;
    e = abs(I-Iprev);
end
I = I*Ncellp*Np;
I = max(I,0);
```

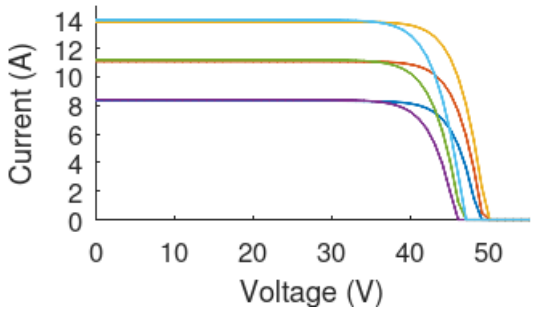


Figure 6.2: Current-voltage plot of the example module

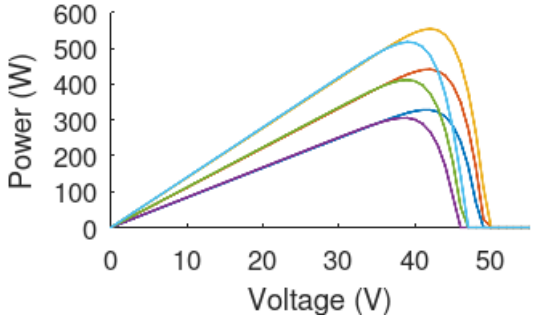


Figure 6.3: Power-voltage plot of the example module

7 (de Blas et al. model)

This model is proposed by Bishop [7], which is published in [15].

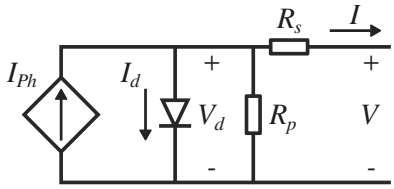


Figure 7.1: One diode R_s and R_p PV cell model

The photovoltaic module is modelled as a current source in parallel to a diode with a series R_s and a shunt resistor R_p (Figure 7.1).

TODO

For a singe cell:

- Current balance (Kirchhoff current law):
- Photocurrent I_{Ph}:
- Shockley diode equation I_d:
- Reverse saturation current I_0:
- Thermal voltage V_t:

Bishop model example

```
function I = fcn(T,G,V)

Gstc = 1e3;
Tstc = 25+273;
DT = T-Tstc;

% Number of modules series / parallel
Ns = 1;
Np = 1;

% Number of cells per module
Ncells = 144/2;
Ncellp = 2;

I = I*Ncellp*Np;
I = max(I,0);
```

[7] M. de Blas, J. Torres, E. Prieto, and A. Garcıa, "Selecting a suitable model for characterizing photovoltaic devices," Renewable Energy, vol. 25, pp. 371–380, Mar. 2002., DOI: 10.1016/s0960-1481(01)00056-8

8 (J.W.Bishop model)

This model is proposed by Bishop [8], which is published in [16].

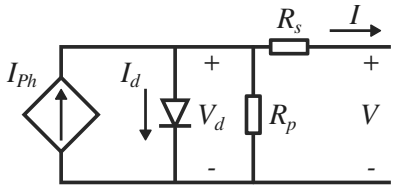


Figure 8.1: One diode R_s and R_p PV cell model

The photovoltaic module is modelled as a current source in parallel to a diode with a series R_s and a shunt resistor R_p (Figure 8.1). It also includes the recombination current and the reverse bias breakdown. See an implementation in the *pvlib* python library[9].

TODO

For a singe cell:

- Current balance (Kirchhoff current law):
- Photocurrent I_{Ph}:
- Shockley diode equation I_d:
- Reverse saturation current I_0:
- Thermal voltage V_t:

Bishop model example

```
function I = fcn(T,G,V)

Gstc = 1e3;
Tstc = 25+273;
DT = T-Tstc;

% Number of modules series / parallel
Ns = 1;
Np = 1;

% Number of cells per module
Ncells = 144/2;
Ncellp = 2;

I = I*Ncellp*Np;
I = max(I,0);
```

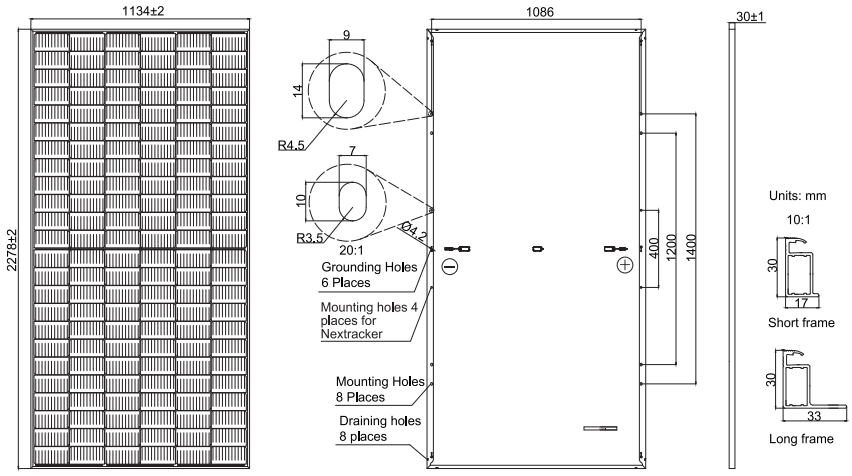
[8] J. Bishop, "Computer simulation of the effects of electrical mismatches in photovoltaic cell interconnection circuits," Solar Cells, vol. 25, pp. 73–89, Oct. 1988., DOI: 10.1016/0379-6787(88)90059-2

[9] pvlib-python.readthedocs.io

JA SOLAR

JAM72S30 530-555/MR Series

MECHANICAL DIAGRAMS



Remark: customized frame color and cable length available upon request

SPECIFICATIONS

Cell	Mono
Weight	27.8kg
Dimensions	2278±2mm×1134±2mm×30±1mm
Cable Cross Section Size	4mm² (IEC), 12 AWG(UL)
No. of cells	144(6×24)
Junction Box	IP68, 3 diodes
Connector	MC4-EVO2/ QC 4.10-351
Cable Length (Including Connector)	Portrait: 200mm(+)/300mm(-); Landscape: 1300mm(+)/1300mm(-)
Packaging Configuration	36pcs/Pallet 720pcs/40HQ Container

ELECTRICAL PARAMETERS AT STC

TYPE	JAM72S30 -530/MR	JAM72S30 -535/MR	JAM72S30 -540/MR	JAM72S30 -545/MR	JAM72S30 -550/MR	JAM72S30 -555/MR
Rated Maximum Power(Pmax) [W]	530	535	540	545	550	555
Open Circuit Voltage(Voc) [V]	49.30	49.45	49.60	49.75	49.90	50.02
Maximum Power Voltage(Vmp) [V]	41.31	41.47	41.64	41.80	41.96	42.11
Short Circuit Current(Isc) [A]	13.72	13.79	13.86	13.93	14.00	14.07
Maximum Power Current(Imp) [A]	12.83	12.90	12.97	13.04	13.11	13.18
Module Efficiency [%]	20.5	20.7	20.9	21.1	21.3	21.5
Power Tolerance	0~+5W					
Temperature Coefficient of Isc(α_Isc)	+0.045%/°C					
Temperature Coefficient of Voc(β_Voc)	-0.275%/°C					
Temperature Coefficient of Pmax(γ_Pmp)	-0.350%/°C					
STC	Irradiance 1000W/m², cell temperature 25°C, AM1.5G					

Remark: Electrical data in this catalog do not refer to a single module and they are not part of the offer.They only serve for comparison among different module types.

ELECTRICAL PARAMETERS AT NOCT

TYPE	JAM72S30 -530/MR	JAM72S30 -535/MR	JAM72S30 -540/MR	JAM72S30 -545/MR	JAM72S30 -550/MR	JAM72S30 -555/MR
Rated Max Power(Pmax) [W]	401	405	408	412	416	420
Open Circuit Voltage(Voc) [V]	46.18	46.31	46.43	46.55	46.68	46.85
Max Power Voltage(Vmp) [V]	38.57	38.78	38.99	39.20	39.43	39.66
Short Circuit Current(Isc) [A]	11.01	11.05	11.09	11.13	11.17	11.21
Max Power Current(Imp) [A]	10.39	10.43	10.47	10.51	10.55	10.59
NOCT	Irradiance 800W/m², ambient temperature 20°C,wind speed 1m/s, AM1.5G					

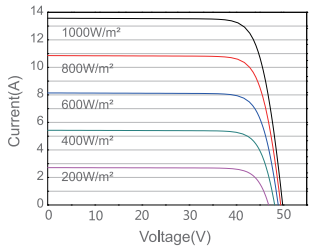
*For NexTracker installations, Maximum Static Load, Front is 1800Pa while Maximum Static Load, Back is 1800Pa.

OPERATING CONDITIONS

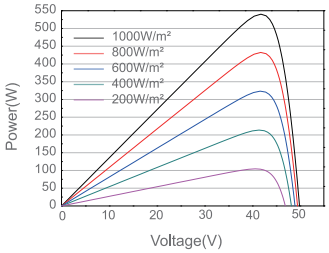
Maximum System Voltage	1000V/1500V DC
Operating Temperature	-40℃~+85℃
Maximum Series Fuse Rating	25A
Maximum Static Load,Front* Maximum Static Load,Back*	5400Pa(112lb/ft²) 2400Pa(50lb/ft²)
NOCT	45±2℃
Safety Class	Class Ⅱ
Fire Performance	UL Type 1

CHARACTERISTICS

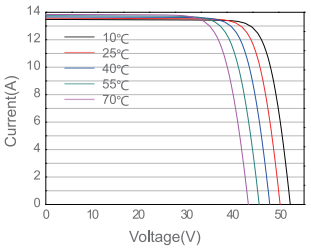
Current-Voltage Curve JAM72S30-540/MR



Power-Voltage Curve JAM72S30-540/MR



Current-Voltage Curve JAM72S30-540/MR



References

[1] IEC, "IEC 61853-1:2011 photovoltaic (pv) module performance testing and energy rating. part 1: Irradiance and temperatureperformance measurements and power rating.," Jan. 2011.

[2] R. Ross Jr, "Interface design considerations for terrestrial solar cell modules," in *12th Photovoltaic specialists conference*, pp. 801–806, 1976.

[3] E. Skoplaki and J. Palyvos, "Operating temperature of photovoltaic modules: A survey of pertinent correlations," *Renewable Energy*, vol. 34, pp. 23–29, Jan. 2009.

[4] G. Valentin, *User manual for PV*SOL version 2.4*. Berlin: Valentin Energiesoftware, 2004.

[5] J. K. D.L. King, W.E. Boyson, "Photovoltaic array performance model," techreport SAND2004-3535, Sandia National Laboratories, Albuquerque, NM, 2004.

[6] D. Faiman, "Assessing the outdoor operating temperature of photovoltaic modules," *Progress in Photovoltaics: Research and Applications*, vol. 16, pp. 307–315, Feb. 2008.

[7] E. Barykina and A. Hammer, "Modeling of photovoltaic module temperature using faiman model: Sensitivity analysis for different climates," *Solar Energy*, vol. 146, pp. 401–416, Apr. 2017.

[8] R. A. Messenger and J. Ventre, *Photovoltaic Systems Engineering*. CRC Press, 2005.

[9] B. Bletterie, R. Bründlinger, H. Häberlin, F. Baumgartner, H. Schmidt, B. Burger, G. Klein, and M. A. Abella, "Redefinition of the european efficiency–finding the compromise between simplicity and accuracy," *Proc. EU PVSEC*, pp. 2735–2742, 2008.

[10] AENOR, "UNE-EN 50530:2011 rendimiento global de los inversores fotovoltaicos," June 2011.

[11] A. Bellini, S. Bifaretti, V. Iacovone, and C. Cornaro, "Simplified model of a photovoltaic module," in *2009 Applied Electronics*, pp. 47–51, 2009.

[12] W. De Soto, S. Klein, and W. Beckman, "Improvement and validation of a model for photovoltaic array performance," *Solar Energy*, vol. 80, pp. 78–88, Jan. 2006.

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